

Theoretical Foundations

Below is a brief summary of the knowledge required to solve the problems that follow.

It is by no means a substitute for theoretical lectures.

Let X be a non-empty set and $*$ a binary operation on it. The operation $*$ on the set X may (but does not have to) satisfy the following properties:

1. *Closure:* $(\forall x, y \in X) x * y \in X$.

This property states that the result of applying the operation to two elements of the set X remains in the set X . For example, the sum of two natural numbers is a natural number, so the addition operation is closed on the set of natural numbers. The difference between two natural numbers is not necessarily a natural number, so the subtraction operation is not closed on the set of natural numbers.

2. *Associativity:* $(\forall x, y, z \in X) (x * y) * z = x * (y * z)$.

3. *Existence of an identity element:*
 $(\exists e \in X) (\forall x \in X) x * e = e * x = x$.

For example, in the set of real numbers, the identity element for the operation of addition is the number 0, while the identity element for the operation of multiplication is the number 1.

4. *Existence of an inverse element:*
 $(\forall x \in X) (\exists x' \in X) x * x' = x' * x = e$.

For example, for the operation of addition on the set of real numbers, the inverse element of an element x is the element $-x$, because $x + (-x) = (-x) + x = 0$, and zero is the identity for addition. The inverse element with respect to the operation of real number multiplication is the reciprocal; therefore, the number 0 has no inverse with respect to real number multiplication.

5. *Commutativity:* $(\forall x, y \in X) x * y = y * x$.

A set together with an operation on it, i.e., the pair $(X, *)$, will be called an algebraic structure. If it satisfies property 1, we say it is a *magma* (or *groupoid*). If it satisfies properties 1 and 2, we say it is a *semigroup*. If properties 1, 2, and 3 are satisfied, it is a *monoid*. If properties 1, 2, 3, and 4 hold, then it is a *group*. Finally, if the commutativity property also holds in a group, then it is an *Abelian*¹ group.

Problems Covered in Class

1. If $A = \{x + y\sqrt{2} \mid x, y \in \mathbb{Q}\}$ and $x * y = x \cdot y$, where $\cdot$ is multiplication of real numbers, examine whether $(A, *)$ is an Abelian group.

1. *Closure.* Let $x + y\sqrt{2}$ and $a + b\sqrt{2}$ be two arbitrary elements of the set A . This implies that $x, y, a, b \in \mathbb{Q}$, that is, they are rational numbers. If the operation $*$ were closed on the set A , it would mean that the result of the operation must be an element of the set A . Let us calculate that result first:

$$\begin{aligned}(x + y\sqrt{2}) * (a + b\sqrt{2}) &= (x + y\sqrt{2}) \cdot (a + b\sqrt{2}) \\ &= xa + xb\sqrt{2} + ya\sqrt{2} + 2yb.\end{aligned}$$

From this form, it is not possible to conclude much, i.e., it is unclear whether the above result of the operation "remains" in the set A . Therefore, we will write it in a more convenient form:

$$(xa + 2yb) + (xb + ya)\sqrt{2}.$$

Now it only remains to state the obvious: the above number is indeed of the desired form ("*something plus something times the square root of two*"), and moreover $xa + 2yb \in \mathbb{Q}$ and $xb + ya \in \mathbb{Q}$, because these are sums and products of rational numbers, so they themselves are rational numbers.

To summarize, we took two arbitrary elements of the set A , applied the operation $*$ to them, and concluded that the result of the operation belongs to that set. Therefore, *the operation $*$ is closed on the set A , i.e., the closure property holds.*

2. *Associativity.* Formally, it would be necessary to take three arbitrary elements of the set A , say $x + y\sqrt{2}$, $a + b\sqrt{2}$, and $\alpha + \beta\sqrt{2}$ (for $x, y, a, b, \alpha, \beta \in \mathbb{Q}$), apply the operation $*$ in two orders according to the definition, and conclude whether the results coincide or not.

However, here we can get away more "cheaply". The operation $*$ is not just any abstract operation, but the well-known multiplication of real numbers, which we know to be associative. Thus, for any three **real** numbers u , v , and w , we know that $(u \cdot v) \cdot w = u \cdot (v \cdot w)$ holds. The elements of the set A are real numbers; not all real numbers, but certainly real numbers. If a property holds for *all* triples of real numbers, then it also holds for our "special" ones.

A formal answer would be: since multiplication of real numbers is associative, and the elements of the set A are real numbers, this property transfers to the set A as well. Therefore, *the associativity property holds.*

5. *Commutativity.* Similar to checking the previous property, the commutativity property is inherited from the set of real numbers, so it **holds** for this set and operation as well.

¹Niels Henrik Abel (1802–1829), Norwegian mathematician

3. *Existence of an identity element.* Since we have the operation of real number multiplication, it is to be expected that the identity element could be² the number 1. The definitional property of the identity certainly holds for it, namely that $1 \cdot (x + y\sqrt{2}) = (x + y\sqrt{2}) \cdot 1 = x + y\sqrt{2}$, where $x + y\sqrt{2}$ is an arbitrary element of A . However, the property of the existence of an identity requires not only that such an ("identity") element exists, but that it exists in the set A . Since $1 = 1 + 0\sqrt{2}$ with $1 \in \mathbb{Q}$ and $0 \in \mathbb{Q}$, we conclude that $1 \in A$, so ***an identity element exists and is equal to the number 1.***

4. *Existence of an inverse.* For this property of existence of an inverse to be satisfied, an arbitrary (hence every) element of the set A must have an inverse. Let $x + y\sqrt{2}$ be (again) an arbitrary element of the set A . Since we are dealing with the operation of real number multiplication, the only possible candidate for the inverse of this element is the number

$$\frac{1}{x + y\sqrt{2}}.$$

Here two questions arise. First, is such a number an element of the set A , i.e., can its expression be "repackaged" into the required form? Second, does this reciprocal value always exist (always = for every element of A)? For the property to hold, an affirmative answer to both questions is necessary. However, the answer to the second question is negative. Namely, we know that $1/0$ is an undefined expression, and zero is indeed an element of the set A , because it can be written as $0 + 0\sqrt{2}$. Therefore, ***there exists (at least) one element³ of the set A that does not have an inverse with respect to the operation $*$. This property does not hold for this algebraic structure.***

We conclude: $(A, *)$ ***is not an Abelian group***, because it is not a group. It is a commutative monoid.

2. If $A_1 = \{x + y\sqrt{2} \mid x, y \in \mathbb{Q}, x^2 + y^2 \neq 0\}$ and $x * y = x \cdot y$, where $\cdot$ is multiplication of real numbers, examine whether $(A_1, *)$ is an Abelian group.

The set A_1 differs from the set A from the previous problem only in the condition $x^2 + y^2 \neq 0$. This condition reads as "among the numbers x and y , at least one is non-zero", because the only way for the sum of the squares of two real numbers to be zero is for both of them to be zero. Thus, the set A_1 is smaller than the set A by exactly

²Even if it turned out that 1 does not belong to the set, that would not be proof that there is no identity. This problem is just one from a class of problems where a well-known set with a well-known operation is taken, where an identity element exists, and then the operation is kept while the original set is reduced, i.e., some elements are removed from it. The main trick with the solutions to such problems, especially regarding the identity and inverse, is to check whether those "typical" identity and inverse elements, inherited from the larger set, "survive" that restriction or fall outside that narrowed set. **It would be good to remember this footnote when we learn matrix calculus.** There we will have famous examples of sets of matrices $S = \left\{ \begin{bmatrix} a & a \\ a & a \end{bmatrix} \mid a \in \mathbb{R} \right\}$ and $G = \left\{ \begin{bmatrix} a & 0 \\ 0 & 0 \end{bmatrix} \mid a \in \mathbb{R} \right\}$, with respect to matrix multiplication, which best illustrate this anomaly to which the footnote refers.

³For property 4 to hold, every element must have an inverse. The negation of the statement "every element has an inverse" ***is not*** "no element has an inverse", but "not all elements have an inverse", which is equivalent to the statement "there exists at least one element that does not have an inverse". That is why it was sufficient to find one element that does not have one, in order to prove that property 4 is not satisfied.

⁴A careful reader might criticize us here for a serious mathematical imprecision. And they would be right. Namely, zero was not removed from the set A , but rather the number $0 + 0\sqrt{2}$. What if zero could be written differently, as some other $a + b\sqrt{2}$, where it is not $a = b = 0$? This is not possible, and the proof is left for independent work to those who can and want more.

⁵If and only if.

one element: $0 + 0\sqrt{2}$, that is, it is obtained from the set A by removing zero.⁴ Now, in the standard way, we examine the properties that an Abelian group must satisfy. We will use the solution of the previous problem wherever possible.

1. *Closure.* Let $x + y\sqrt{2}$ and $a + b\sqrt{2}$ be arbitrary elements of the set A_1 . Thus, among other things, neither of them is zero. Since they are also elements of the set A , their product is certainly in A because $*$ is closed on the set A . The only thing that could cause a problem is if their product equals zero, thereby "jumping out" of A_1 . However, that is not possible because the product of two numbers is zero iff⁵ one of them is zero. In our case, by assumption, that is not the case. Therefore, ***closure holds.***

2. *Associativity.* ***Holds*** because it is inherited from the broader set A (to which it was inherited from the broader set $\mathbb{R}$).

5. *Commutativity.* ***Holds*** because it is inherited from the broader set A (to which it was inherited from the broader set $\mathbb{R}$).

3. *Existence of an identity element.* The number 1 is again a candidate, because the operation remained unchanged. Here, we only need to check that the number 1 did not accidentally "fall out of the set" during the restriction of A to A_1 . However, only zero was removed, and one is not zero, so $1 \in A_1$; thus, here too, ***an identity element exists and is equal to the number 1.***

4. *Existence of an inverse.* Here is where things got stuck in the previous problem. So, let $x + y\sqrt{2}$ be an arbitrary element of the set A_1 . Since it can no longer be equal to zero, and since we are dealing with the operation of real number multiplication, a good candidate for its inverse is the real number

$$\frac{1}{x + y\sqrt{2}},$$

which now definitely exists. It certainly gives one when multiplied by $x + y\sqrt{2}$, but we need to check whether it belongs to the set A_1 . Let us transform it into a more favorable form:

$$\begin{aligned} \frac{1}{x + y\sqrt{2}} &= \frac{1}{x + y\sqrt{2}} \cdot \frac{x - y\sqrt{2}}{x - y\sqrt{2}} = \frac{x - y\sqrt{2}}{x^2 - 2y^2} \\ &= \frac{x}{x^2 - 2y^2} + \frac{-y}{x^2 - 2y^2} \sqrt{2}. \end{aligned}$$

We have successfully written the potential inverse in the appropriate form: "something plus something

times square root of two". But we have a few more small details to fix. We know that, by the definition of the set A_1 , $x + y\sqrt{2}$ is not equal to zero. However, in addition to it, division by the numbers $x - y\sqrt{2}$ and $x^2 - 2y^2$ appears. It is clear that $x - y\sqrt{2} \neq 0$ for practically the same reasons as $x + y\sqrt{2}$, since $x^2 + (-y)^2 = x^2 + y^2$, which we know is non-zero. Then $x^2 - 2y^2 \neq 0$ as well, because it is equal to the product of two numbers that are not equal to zero. Furthermore,

$$\frac{x}{x^2 - 2y^2} \in \mathbb{Q} \quad \text{and} \quad \frac{-y}{x^2 - 2y^2} \in \mathbb{Q},$$

because they represent sums, products, and quotients of rational numbers. Further,

$$\left(\frac{x}{x^2 - 2y^2}\right)^2 + \left(\frac{-y}{x^2 - 2y^2}\right)^2 = \frac{x^2 + y^2}{(x^2 - 2y^2)^2},$$

which is not equal to zero because, by the definition of the set A_1 , $x^2 + y^2 \neq 0$. Therefore,

$$\frac{1}{x + y\sqrt{2}} \in A_1.$$

In this problem, all necessary conditions are satisfied. Thus, $(A_1, *)$ **is an Abelian group**.

3. If $M = \mathbb{Q} \setminus \{-1\}$ and $a * b = a + b + ab$, examine whether $(M, *)$ is an Abelian group.

We solve the problem by examining the required properties, as with the previous and every subsequent problem. This problem is slightly more difficult than the previous one because the given operation is not one of the familiar ones, so we will have to examine all properties "by hand", almost without referencing properties of previously known operations.

1. *Closure.* Let a and b be arbitrary elements of the set M . This means that they are rational numbers and that neither of them is equal to -1 . Certainly $a * b = a + b + ab$ is a rational number, as the sum and product of such numbers. Closure "breaks" if it can be equal to -1 for some a and b in M . Suppose that this can happen. That would mean $a + b + ab = -1$, and with a little algebraic rearrangement, this expression becomes

$$(a + 1)(b + 1) = 0.$$

So, if we assume that the operation is not closed on the set, we obtain that the above product is zero for some $a, b \in M$, which is equivalent to $a = -1$ or $b = -1$, which is impossible because they were

chosen from M , and -1 does not belong there. We have derived a contradiction,⁶ so we conclude that $a * b = -1$ cannot happen for any $a, b \in M$, and therefore **the operation $*$ is closed on the set M** .

2. *Associativity.* We check this property by definition, without relying on any shortcuts. Let a, b , and c be three arbitrary elements of the set M . Using the commutativity of addition and multiplication of real numbers and the distributivity of multiplication over addition, we have

$$\begin{aligned} (a * b) * c &= (a + b + ab) * c \\ &= (a + b + ab) + c + (a + b + ab)c \\ &= a + b + c + ab + ac + bc + abc \end{aligned} \quad (1)$$

and

$$\begin{aligned} a * (b * c) &= a * (b + c + bc) \\ &= a + (b + c + bc) + a(b + c + bc) \\ &= a + b + c + ab + ac + bc + abc. \end{aligned} \quad (2)$$

Equalities (1) and (2) give us that $(a * b) * c = a * (b * c)$, so **the associativity property holds**.

5. *Commutativity.* For arbitrary $a, b \in M$, we have

$$a * b = a + b + ab \quad \text{and} \quad b * a = b + a + ba,$$

which are equal expressions, because addition and multiplication of rational (and real) numbers are commutative. Therefore, commutativity holds.⁷

3. *Existence of an identity element.* The problem asks us to examine whether $(M, *)$ is an Abelian group, so we are not sure whether to look for an identity element or to prove that it does not exist. Be that as it may, we will engage in the procedure of finding it, and if it does not exist, that will reveal itself along the way.

We are looking, then, for an element $e \in M$ for which $e * a = a * e = a$ holds for all $a \in M$. Since we have shown that commutativity holds, it suffices to find an element $e \in M$ such that $a * e = a$ for all $a \in M$. Then $a + e + ae = a$ must hold, i.e., $e(a + 1) = 0$. Since $a \in M$, it is not equal to -1 , so $a + 1$ is not equal to zero. From this, we conclude that if an identity exists, it must be $e = 0$. Since $0 \in \mathbb{Q}$ and $0 \neq -1$, it indeed is an identity element. We conclude that **an identity element exists and is equal to 0**.

4. *Existence of an inverse.* This property will be satisfied if for an arbitrary element $a \in M$ there exists an element $a' \in M$ such that $a * a' = 0$ ($a' * a = 0$ then follows from commutativity). Therefore, let $a \in M$

⁶We have just encountered the proof method known as *Proof by Contradiction* (Lat. *Reductio ad absurdum*). The essence of this method is that, if we wish to prove a statement, we assume that its negation holds. Then we correctly derive a series of conclusions, and at the end of that series, we reach a conclusion that we know is untrue (something like $0 = 1$, or, in this problem, that $a = -1$ or $b = -1$, which we previously know is impossible). If our (opposite) assumption led us to a false conclusion through valid logical reasoning, then it must be that our assumption is incorrect, so we immediately know that its negation must hold: namely, the initial statement we wanted to prove.

⁷Here, and in problems in general, it is convenient to check commutativity at the very beginning rather than at the end. If a problem asks whether the structure is an Abelian group and commutativity fails, we have saved ourselves work because we immediately conclude that it is not. If, on the other hand, commutativity holds, then when checking the existence of an identity and an inverse, we do not have to check whether they satisfy both defining equations, but only one, since the other follows from commutativity. This saves us about 50% of the time when checking those items.

be an arbitrary element. We look for its inverse a' satisfying $a * a' = 0$, i.e.,

$$a + a' + aa' = 0,$$

or $a'(1 + a) = -a$. Since $a \in M$, we can divide by $1 + a$, obtaining

$$a' = \frac{-a}{a+1}.$$

The a' defined in this way satisfies the defining property of an inverse element; it remains only to check whether it belongs to the set M . It is clearly a rational number, and if it were equal to -1 , then we would have $-a/(a+1) = -1$, that is, $a = a+1$, i.e., $0 = 1$. This is, of course, a contradiction, so we conclude that a' indeed belongs to the set M , thereby establishing that **every element of the set M has an inverse element in the set M** .

Finally, we conclude that since all necessary conditions are satisfied, $(M, *)$ **is an Abelian group**.

4. If $S = \{(a, b) \mid a, b \in \mathbb{Q}, a \neq 0\}$ and $(a, b) * (c, d) = (ac, bc + c + d)$, examine whether $(S, *)$ is a group. If it is, examine whether it is Abelian.

1. *Closure*. For arbitrary $(a, b), (c, d) \in S$, by definition:

$$(a, b) * (c, d) = (ac, bc + c + d).$$

We have that $ac \neq 0$, because $a \neq 0$ and $c \neq 0$, since the pairs (a, b) and (c, d) were chosen from S , where only those pairs live whose first component is non-zero. Of course, $ac \in \mathbb{Q}$ and $bc + c + d \in \mathbb{Q}$, because $a, b, c, d \in \mathbb{Q}$ for the same reason. Therefore, **the operation $*$ is closed on the set S** .

2. *Associativity*. Let $(a, b), (c, d), (f, g) \in S$ be arbitrary elements. We have

$$\begin{aligned} ((a, b) * (c, d)) * (f, g) &= (ac, bc + c + d) * (f, g) \\ &= (acf, (bc + c + d)f + f + g) \\ &= (acf, bcf + cf + df + f + g), \end{aligned}$$

and

$$\begin{aligned} (a, b) * ((c, d) * (f, g)) &= (a, b) * (cf, df + f + g) \\ &= (acf, bcf + cf + df + f + g). \end{aligned}$$

We see⁸ that the above two expressions are equal, so we conclude that **associativity holds**.

5. *Commutativity*. The definition of the operation $*$ tells us that

$$\begin{aligned} (a, b) * (c, d) &= (ac, bc + c + d) \\ (c, d) * (a, b) &= (ca, da + a + b). \end{aligned}$$

For commutativity to hold, the above two ordered pairs must be equal. Their first components are equal because $ac = ca$; however, one can sense that

the second components are not. We will find a counterexample:

$$(1, 2) * (3, 4) = (1 \cdot 3, 2 \cdot 3 + 3 + 4) = (3, 13)$$

$$(3, 4) * (1, 2) = (3 \cdot 1, 4 \cdot 1 + 1 + 2) = (3, 7).$$

Since $(3, 13) \neq (3, 7)$, and both pairs obviously belong to the set S , we conclude that for the operation $*$ on the set S , **commutativity does not hold**. $(S, *)$ is not an Abelian group, but it can still be a group, so we continue with the verification.

3. *Existence of an identity element*. An ordered pair $(e_1, e_2) \in S$ will be an identity element in our structure if for every element $(a, b) \in S$ it holds that

$$(e_1, e_2) * (a, b) = (a, b) * (e_1, e_2) = (a, b).$$

We will search for $(e_1, e_2) \in S$ satisfying $(a, b) * (e_1, e_2) = (a, b)$, and if we find it, we will check whether the first equality holds for it. Thus, it would have to be

$$(a, b) * (e_1, e_2) = (ae_1, be_1 + e_1 + e_2) = (a, b),$$

that is, $ae_1 = a$, $be_1 + e_1 + e_2 = b$. From this system, we find that $e_1 = 1$ and $b \cdot 1 + 1 + e_2 = b$, which yields $e_2 = -1$. The identity element could be $(e_1, e_2) = (1, -1)$, which also belongs to the set S . Let us check from the other side as well:

$$(e_1, e_2) * (a, b) = (1, -1) * (a, b) = (a, -a + a + b) = (a, b).$$

Therefore, $(1, -1) \in S$, so **an identity element exists and is equal to the ordered pair $(1, -1)$** .

4. *Existence of an inverse*. Let $(a, b) \in S$ be an arbitrary element. We look for its inverse element $(a', b') \in S$ satisfying

$$(a', b') * (a, b) = (a, b) * (a', b') = (1, -1).$$

Similar to finding the identity, we will look for $(a', b') \in S$ such that it satisfies $(a, b) * (a', b') = (1, -1)$, and if we find it, we will check the second equality for that element. It should be

$$(a, b) * (a', b') = (aa', ba' + a' + b') = (1, -1),$$

that is, $aa' = 1$ and $ba' + a' + b' = -1$. Since $a \neq 0$ because $(a, b) \in S$, we immediately calculate that $a' = 1/a$. Substituting into the second equation, we derive:

$$\begin{aligned} \frac{b}{a} + \frac{1}{a} + b' &= -1 \\ b' &= \frac{-1 - a - b}{a}, \end{aligned}$$

which is certainly a rational number. Thus, the ordered pair

$$(a', b') = \left(\frac{1}{a}, \frac{-1 - a - b}{a} \right) \in S$$

is a good candidate for the inverse of the element (a, b) . We check it in the other order as well:

$$(a', b') * (a, b) = \left(\frac{1}{a}, \frac{-1 - a - b}{a} \right) * (a, b)$$

⁸Implicitly using the commutativity of addition and multiplication of real numbers and the distributivity of multiplication over addition.

$$\begin{aligned}
&= \left(\frac{1}{a} \cdot a, \frac{-1-a-b}{a} \cdot a + a + b \right) \\
&= (1, -1-a-b+a+b) \\
&= (1, -1).
\end{aligned}$$

Thus, an arbitrary element $(a, b) \in S$ has an inverse element in S .

All necessary criteria except commutativity are satisfied, so we conclude that $(S, *)$ **is a group, but not an Abelian group**.

5. If $K = \{(a, b) \mid a, b \in \mathbb{Z}\}$ and $(a, b) * (c, d) = (a + 2^b c, b + d)$, examine whether $(K, *)$ is an Abelian group.

1. *Closure.* Take $(0, -1), (1, 1) \in K$. We have

$$(0, -1) * (1, 1) = (0 + 2^{-1} \cdot 1, -1 + 1) = \left(\frac{1}{2}, 0\right),$$

which does not belong to the set K , because $1/2 \notin \mathbb{Z}$. Therefore, the operation $*$ is not even closed on the set K , so $(K, *)$ **is not an Abelian group**.

Additional Solved Problems

6. Let $G = \mathbb{Q} \setminus \{-\frac{1}{2024}\}$ and let the operation on G be defined as

$$x * y = x + y + 2024xy, \quad x, y \in G.$$

Examine whether $(G, *)$ is a group. Is $(G, *)$ an Abelian group?

1. *Closure.* Suppose that $x * y = -1/2024$ could hold for some $x, y \in G$. Then we would have

$$x + y + 2024xy = -\frac{1}{2024}.$$

Equivalently:

$$\begin{aligned}
2024x + 2024y + 2024^2 xy &= 1 \\
1 + 2024x + 2024y(1 + 2024x) &= 0 \\
(1 + 2024x)(1 + 2024y) &= 0.
\end{aligned}$$

From here, it would follow that $1 + 2024x = 0$ or $1 + 2024y = 0$, that is, $x = -1/2024$ or $y = -1/2024$. This is a contradiction with $x, y \in G$. Therefore, **the operation $*$ is closed on the set G** .

2. *Associativity.* Let $x, y, z \in G$ be arbitrary elements. Taking into account the commutativity and associativity of multiplication and addition of real numbers, as well as the distributivity of multiplication over addition, we have

$$\begin{aligned}
&(x * y) * z \\
&= (x + y + 2024xy) * z \\
&= (x + y + 2024xy) + z + 2024(x + y + 2024xy)z
\end{aligned}$$

$$= x + y + z + 2024xy + 2024xz + 2024yz + 2024^2xyz$$

and

$$\begin{aligned}
&x * (y * z) \\
&= x * (y + z + 2024yz) \\
&= x + (y + z + 2024yz) + 2024x(y + z + 2024yz) \\
&= x + y + z + 2024xy + 2024xz + 2024yz + 2024^2xyz.
\end{aligned}$$

The above two expressions are equal, so **associativity holds**.

5. *Commutativity.* For $x, y \in G$ we have

$$\begin{aligned}
x * y &= x + y + 2024xy \\
y * x &= y + x + 2024yx,
\end{aligned}$$

which are equal based on properties of real numbers. Therefore, **commutativity holds**.

3. *Existence of an identity element.* We look for $e \in G$ such that⁹ for every $x \in G$, $x * e = x$ is satisfied, i.e.,

$$x + e + 2024xe = x,$$

which reduces to $e(1 + 2024x) = 0$. Since $1 + 2024x \neq 0$ (because $x \in G$), it must be $e = 0$. Since $0 \in \mathbb{Q}$ and $0 \neq -1/2024$, we conclude that $0 \in G$, so **an identity element exists and is equal to 0**.

4. *Existence of an inverse.* Let $x \in G$ be an arbitrary element. We look for an element $x' \in G$ such that¹⁰ $x * x' = 0$, i.e.,

$$x + x' + 2024xx' = 0,$$

which reduces to $x'(1 + 2024x) = -x$, that is,

$$x' = \frac{-x}{1 + 2024x}.$$

Division is justified because $1 + 2024x \neq 0$ since $x \in G$. Such x' is obviously rational, and if it were $x' = -1/2024$, we would have

$$\begin{aligned}
\frac{-x}{1 + 2024x} &= -\frac{1}{2024} \iff 2024x = 1 + 2024x \\
&\iff 0 = 1,
\end{aligned}$$

which is, of course, a contradiction. Thus, x' satisfies the required property, and moreover $x' \in G$. It follows that **every element from G has an inverse in G with respect to the operation $*$** .

We conclude that $(G, *)$ **is an Abelian group**, because it satisfies all necessary criteria.

7. Let $G = (-1, 1)$ and let the operation $*$ be defined as:

$$a * b = \frac{a + b}{1 + ab}, \quad a, b \in G.$$

Examine whether $(G, *)$ is a group. Is $(G, *)$ an Abelian group?

⁹ $x * e = e * x$ holds due to commutativity, so it is sufficient to check only one side.

¹⁰See previous footnote.

1. *Closure.* Let $a, b \in G$ be arbitrary elements. Since they are numbers strictly between -1 and 1 , so is ab , so $1 + ab$ is a strictly positive number. This will be important to us because we will be able to multiply and divide inequalities by it.

Suppose $a * b \geq 1$, i.e.,

$$\frac{a+b}{1+ab} \geq 1.$$

Equivalently:

$$\begin{aligned} a+b &\geq 1+ab \\ a-1+b-ab &\geq 0 \\ (a-1)-b(a-1) &\geq 0 \\ (a-1)(1-b) &\geq 0 \\ (a-1)(b-1) &\leq 0. \end{aligned}$$

However, since $a, b \in G = (-1, 1)$, then $a-1, b-1 \in (-2, 0)$, so both are strictly negative numbers; then their product is a strictly positive number, so we arrive at a contradiction. Thus, $a * b \geq 1$ cannot hold for $a, b \in G$.

Now suppose that $a * b \leq -1$ for $a, b \in G$. Equivalently:

$$\begin{aligned} \frac{a+b}{1+ab} &\leq -1 \\ a+b &\leq -1-ab \\ 1+a+b+ab &\leq 0 \\ (a+1)(b+1) &\leq 0. \end{aligned}$$

Similarly, since $a, b \in G = (-1, 1)$, then $a+1, b+1 \in (0, 2)$, i.e., they are strictly positive numbers, so their product is strictly positive. Again a contradiction. Thus, for $a, b \in G$, it must be that $a * b \in (-1, 1) = G$, which means that **closure holds**.

2. *Associativity.* Let $a, b, c \in G$ be arbitrary. We calculate that

$$\begin{aligned} (a * b) * c &= \left(\frac{a+b}{1+ab} \right) * c = \frac{\frac{a+b}{1+ab} + c}{1 + \frac{a+b}{1+ab} \cdot c} \\ &= \frac{\frac{a+b+c(1+ab)}{1+ab}}{\frac{1+ab+(a+b)c}{1+ab}} = \frac{a+b+c+abc}{1+ab+ac+bc} \end{aligned}$$

and

$$\begin{aligned} a * (b * c) &= a * \left(\frac{b+c}{1+bc} \right) = \frac{a + \frac{b+c}{1+bc}}{1 + a \cdot \frac{b+c}{1+bc}} \\ &= \frac{\frac{a(1+bc)+b+c}{1+bc}}{\frac{1+bc+a(b+c)}{1+bc}} = \frac{a+abc+b+c}{1+bc+ab+ac}. \end{aligned}$$

The above two final expressions are equal, thanks to the properties of real numbers. Thus, in this structure, **associativity holds**.

5. *Commutativity.* It is clear that for $a, b \in G$ the expressions

$$a * b = \frac{a+b}{1+ab} \quad \text{and} \quad b * a = \frac{b+a}{1+ba}$$

are equal, because addition and multiplication of real numbers are commutative operations. Therefore, **commutativity holds**.

3. *Existence of an identity element.* Due to commutativity, it is sufficient to find $e \in G$ such that for all $a \in G$, $a * e = a$ is satisfied, i.e.,

$$\frac{a+e}{1+ae} = a,$$

which with quick computation reduces to $a + e = a + a^2e$, i.e., $e(1 - a^2) = 0$. Equivalently,

$$e(1-a)(1+a) = 0.$$

Since $a \in G = (-1, 1)$, we have $a \neq -1$ and $a \neq 1$, so $(1-a)(1+a) \neq 0$. It follows that $e = 0$, which certainly belongs to G . Thus, **an identity element exists and is equal to 0**.

4. *Existence of an inverse.* For $x \in G$, due to commutativity, it is sufficient to look for $x' \in G$ for which $x * x' = 0$, i.e.,

$$\frac{x+x'}{1+xx'} = 0.$$

It must be $x + x' = 0$, i.e., $x' = -x$. Such an x' indeed belongs to G , because G is symmetric about zero. Thus, **every element from G has, with respect to the operation $*$, an inverse in G** .

All criteria are satisfied, so $(G, *)$ **is an Abelian group**.

Problems for Independent Work

8. On the set $G = (\mathbb{R} \setminus \{0\}) \times \mathbb{R}$ we define the binary operation $*$ as

$$(a, b) * (x, y) = (ax, a^2y + b), \quad (a, b), (x, y) \in G.$$

Prove that $(G, *)$ is a group. Is this group Abelian?

9. Let $G = \{(a, b) \mid a, b \in \mathbb{Q} \wedge a^2 + b^2 \neq 0\}$ and let the operation $*$ be defined as:

$$(a, b) * (c, d) = (ad + bc, bd - 2019ac), \quad (a, b), (c, d) \in G.$$

Examine whether $(G, *)$ is a group. Is $(G, *)$ an Abelian group?

10. Let $G = (-3, +\infty)$, and let the operation $*$ be defined as:

$$x * y = xy + 3x + 3y + 6, \quad x, y \in G.$$

Examine whether $(G, *)$ is a group. Is $(G, *)$ an Abelian group?